TEST PLAN DOCUMENT

Air Ticket Reservation System

Project Code:	
Internal Advisor:	
External Advisor:	
Project Manager:	Muhammad Ehtisham
Project Team:	Muhammad Ehtisham, Malik Muhammad Umair, Ali Zain
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Project Manager's Signature

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Definition of Terms, Acronyms and Abbreviations

This section provides definitions of all terms, acronyms, and abbreviations required to correctly interpret this document.

Term / Acronym	Description
ATRS	Air Ticket Reservation System
TP	Test Plan
TC	Test Case
SRS	Software Requirements Specification
UC	Use Case
GUI	Graphical User Interface
QA	Quality Assurance
DIA	Dubai International Airport
SIA	Sharjah International Airport
AIA	Abu Dhabi International Airport / Jeddah Airport
KIA	King Khalid International Airport (Riyadh)
FIA	King Fahd International Airport (Dammam) / Faisalabad International Airport
CNIC	Computerized National Identity Card
PIA	Pakistan International Airlines
LHE	Lahore (Allama Iqbal International Airport)
ISB	Islamabad International Airport
GMT	Greenwich Mean Time
Pass	Test case executed and result matches expected
Fail	Test case executed but result does not match expected
N/E	Not Executed – test not yet run

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1. Introduction

The Air Ticket Reservation System (ATRS) is a console-based C++ application that enables users to search, book, and manage air tickets for flights originating from three Pakistani airports (Lahore, Faisalabad, Islamabad) to two international destinations (UAE and Saudi Arabia). The system supports user account management, flight search, seat selection, multi-passenger booking, multi-currency payment processing, and flight cancellation.

1.1 Purpose of Document

This document defines the test plan and test cases for the Air Ticket Reservation System. It is intended to guide the QA team, developers, and project stakeholders in understanding what will be tested, how it will be tested, and what constitutes acceptable test outcomes. The document ensures complete coverage of both functional and non-functional requirements as defined in the SRS.

Intended audience: QA Engineers, Developers, Project Manager, Internal Advisor.

1.2 Project Overview

The ATRS is a C++ object-oriented application implementing a class hierarchy: Person → User_authentication → Booking → Payment → Receipt. It manages user accounts in memory, provides flight data for 54 route/type combinations across 3 departure airports, manages 50-seat aircraft (Economy: 1-30, Business: 31-40, First Class: 41-50), processes payments in 5 currencies, and generates booking receipts. The system is designed for academic submission to FAST-NU and demonstrates core software engineering principles.

2. Scope of Testing

The following areas ARE in scope for testing:

- User authentication: sign-up, login, view/edit account details, forgot password
- Flight search: all departure airport × arrival country × arrival airport × date × flight type combinations
- Date validation: boundary dates, past dates, invalid month/day values
- Booking process: passenger count/type constraints, class selection, baggage weight validation, fare calculation
- Seat management: booking available seats, preventing double-booking, seat cancellation, class-range enforcement
- Payment: all 5 currencies, all 3 payment methods, sufficient/insufficient payment, card/bank input validation
- Receipt generation: completeness of passenger and flight data on receipt
- Flight cancellation: CNIC matching, seat release, confirmation flow
- Input validation: non-numeric inputs for numeric fields, empty strings, out-of-range menu selections

The following areas are OUT of scope:

- Persistence across application restarts (data is stored in-memory only)
- Graphical User Interface (system is console-based)
- Network connectivity or real airline API integration
- Database-level concurrency testing (single-user console application)
- Performance load testing (single-user architecture)

3. Test Plan Strategy

The objective of this test plan is to ensure complete validation of all functional requirements defined in the ATRS SRS v2.0, identify defects in edge cases and boundary conditions derived from source code analysis, and confirm that the system handles invalid inputs gracefully without crashing.

3.1 Unit Testing

Definition: Unit testing validates individual functions and methods in isolation.

Participants: Developers (Muhammad Ehtisham, Malik Muhammad Umair, Ali Zain)

Methodology:

- Each class method (check_valid_name, check_valid_date, fare_calculation, currency_convertor, payment_checker) will be tested independently.
- Input/output boundary conditions will be verified for each function.
- Developers will conduct initial unit tests as modules are completed.

3.2 Integration Testing

Definition: Integration testing verifies that individual modules interact correctly when combined.

Participants: QA Team / Developers

Methodology:

- Test the flow: Search_flight → Booking → Payment → Receipt as an end-to-end chain.
- Verify that flight IDs returned by Search_flight are correctly passed to Seat_management and Payment modules.
- Test that User_authentication credentials propagate correctly into the Booking module.
- Validate that seat state persists correctly from booking through cancellation.

3.3 System Testing

Definition: System testing validates the complete, integrated system against all requirements.

Participants: QA Team Lead

Methodology:

- Execute all test cases in Sections 9.1 through 9.7 using the compiled ATRS binary.
- Test all 7 main menu options as complete user journeys.
- Verify system behaviour for all positive, negative, boundary, and edge cases.
- Record actual results and compare against expected results.
- Document defects with severity and priority ratings.

4. Test Environment

Component	Specification
Operating System	Windows 10/11 or Ubuntu Linux 20.04+
Compiler	g++ (GCC 9.0+) with C++11/14/17 standard
Compile Command	g++ -std=c++17 -o atrs main.cpp
Hardware	Minimum: 4GB RAM, 1GHz CPU, 100MB free disk
IDE/Editor	Visual Studio Code, Code::Blocks, or terminal
Test Data	Manually entered per test case; no external files required
Test Documentation	This document (ATRS-2025-TP)

5. Schedule

No.	Phase / Activity	Start Date	End Date	Responsible
1	Requirement Review & Test Planning	Week 1	Week 1	QA Lead
2	Test Case Design & Documentation	Week 2	Week 2	QA Team
3	Unit Testing	Week 3	Week 3	Developers
4	Integration Testing	Week 4	Week 4	QA Team
5	System Testing	Week 5	Week 6	QA Team
6	User Acceptance Testing (UAT)	Week 7	Week 7	QA Lead / Client
7	Defect Fixing & Retesting	Week 8	Week 8	Developers / QA
8	Final Test Summary Report	Week 9	Week 9	QA Lead

6. Control Activities

The following control procedures govern this test effort:

- All test cases must be reviewed and signed off by the QA Lead before execution begins.
- Defects found during testing will be logged with: Defect ID, Severity (Critical/Major/Minor), Description, Steps to Reproduce, Expected vs Actual Result, and Status.
- Any defect with Critical or Major severity must be fixed and retested before proceeding to the next testing phase.
- Test results will be documented in a Test Summary Report at the end of each testing phase.
- Changes to requirements after test case freeze must go through change control and trigger test case updates.

7. Functions to be Tested

The following functional areas will be covered by this test plan:

- User Sign-Up: name validation, CNIC/passport/phone input, email entry, password criteria enforcement
- User Login: credential matching, error messaging for invalid credentials
- View Account Details: correct display of stored user information
- Edit Account Details: updating name, phone, and password
- Forgot Password: email+phone verification, 4-digit code validation, new password setting
- Flight Search: departure airport selection, arrival country and airport input, date validation, one-way vs return
- Flight Information Display: flight ID, name, type, from, to, date, price, baggage allowance, times, refund, upgrade
- Passenger Management: count limits (1-5), adult/child/infant rules
- Seat Selection: class selection, seat number booking by class range, availability display
- Seat Cancellation: CNIC matching, ownership check, double-cancellation prevention
- Baggage Weight: boundary validation (0-20kg), fare calculation
- Fare Calculation: UAE (100000 base) and SA (150000 base) with baggage surcharge
- Currency Conversion: PKR equivalent calculation for USD, EUR, GBP, AED, SAR
- Payment Processing: debit card, credit card, online banking flows
- Payment Validation: sufficient vs insufficient amount, change calculation
- Receipt Generation: complete passenger and flight detail display
- Flight Cancellation: CNIC matching, booking removal, seat release
- Input Validation: cin error handling, range checks, string validation
- Menu Navigation: all 9 main menu options including exit

8. Functions Not to be Tested

- Data persistence between application sessions (not implemented; data is in-memory only)
- Actual SMS/OTP delivery for forgot password (simulated by user entering any 4-digit code)
- Real payment gateway integration (card/bank details are accepted but not validated against a real gateway)
- Graphical or web-based user interface (console application only)
- Concurrent multi-user access (single-process, single-user design)
- Real-time flight availability from external airline APIs
- File I/O or database operations (not implemented)
- Network security or HTTPS (console application; no network layer)

9. Test Case Design and Description

Each test case below follows the standard template format including: Test Case ID, Test Scenario, Preconditions, Test Steps, Test Data, Expected Result, Actual Result (to be filled during execution), and Status.

Test Case ID:	[See tables below]	**Version:**	1.0	**Testing Team Lead:**	Muhammad Ehtisham
Test Case Version:	1.0	**Reviewed By:**	Internal Advisor	**Relation to Use Cases:**	UC-01 to UC-12
Test Date:	March 30, 2026	**Use Case Reference(s):**	UC-01 through UC-12		
Revision History:	Initial version, no prior test case identity				
Objective:	Validate all functional requirements of the Air Ticket Reservation System through structured test cases covering positive, negative, boundary, and edge scenarios.				
Product/Ver/Module:	Air Ticket Reservation System v1.0 – All Modules (User Auth, Flight Search, Seat Mgmt, Booking, Payment, Receipt, Cancellation)				
Environment:	g++ compiler, Windows 10/Ubuntu 20.04, console terminal				
Assumptions:	System compiled successfully; data is in-memory; tests run in single session; forgot password OTP simulated by any 4-digit input.				
Pre-Requisite:	ATRS binary compiled from source. For booking tests: at least one user account signed up.				
Step No.	**Execution Description (Events being tested)**	**Procedure Result (Software response)**			

See Section 9 test case tables below (Steps column)	Detailed step-by-step execution is provided per test case.	Expected results are defined per test case. Actual results and Pass/Fail status to be completed during test execution.
Comments:	Passed Failed Not Executed	

9.1 Module 1 – User Authentication

This module covers all aspects of user account management: registration, login, profile viewing, profile editing, and password recovery.

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-UA-001	Successful User Sign-Up (Positive)	No account exists for the email being registered.	1. Launch system 2. Select option 1 (SignUp) 3. Enter valid first name: 'Ali' 4. Enter valid last name: 'Khan' 5. Enter valid CNIC: 3520212345671 6. Enter valid passport: 12345678 7. Enter phone: 03001234567 8. Enter email: ali.khan@test.com 9. Enter password: Ali@12345	Name: Ali Khan CNIC: 3520212345671 Passport: 12345678 Phone: 03001234567 Email: ali.khan@test.com Password: Ali@12345	System displays 'Account Created successfully!' and stores the user record.		
TC-UA-002	Sign-Up with Name Containing Digits (Negative)	No prior account.	1. Launch system 2. Select option 1 (SignUp) 3. Enter first name: 'Al1' 4. Observe system response	First Name: Al1	System displays 'Numbers and special characters are not allowed!' and re-prompts for name.		
TC-UA-003	Sign-Up with Name Containing Special Characters (Negative)	No prior account.	1. Select SignUp 2. Enter first name: 'Al!@' 3. Observe response	First Name: Al!@	System displays 'Numbers and special characters are not allowed!' and re-prompts.		
TC-UA-004	Sign-Up with Weak Password (< 8 chars) (Negative)	Valid name and CNIC entered.	1. Proceed through name/CNIC entry 2. Enter password: 'Ali@1' (5 chars) 3. Observe response	Password: Ali@1	System displays 'Password must have atleast 8 characters long!' and re-prompts.		
TC-UA-005	Sign-Up with Password Missing Uppercase (Negative)	Valid name and CNIC entered.	1. Proceed through name/CNIC entry 2. Enter password: ali@12345 (no uppercase) 3. Observe response	Password: ali@12345	System displays 'Password does not match with criteria Try Again!' and re-prompts.		

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-UA-006	Sign-Up with Password Missing Special Character (Negative)	Valid name and CNIC entered.	1. Proceed to password entry 2. Enter password: Ali12345 (no special char) 3. Observe response	Password: Ali12345	System displays 'Password does not match with criteria Try Again!' and re-prompts.		
TC-UA-007	Sign-Up with Password Missing Digit (Negative)	Valid name and CNIC entered.	1. Proceed to password entry 2. Enter password: Ali@abcd (no digit) 3. Observe response	Password: Ali@abcd	System displays 'Password does not match with criteria Try Again!' and re-prompts.		
TC-UA-008	Sign-Up with Password Boundary: Exactly 8 Valid Characters (Boundary)	Valid name and CNIC entered.	1. Proceed to password entry 2. Enter password: Ali@1234 (exactly 8 chars, all criteria met) 3. Observe response	Password: Ali@1234	System accepts the password and creates the account successfully.		
TC-UA-009	Successful Login (Positive)	A user account with email 'ali.khan@test.com' and password 'Ali@12345' exists.	1. Select option 2 (Login) 2. Enter email: ali.khan@test.com 3. Enter password: Ali@12345 4. Observe response	Email: ali.khan@test.com Password: Ali@12345	System displays 'Login successful'.		
TC-UA-010	Login with Wrong Password (Negative)	Account with ali.khan@test.com exists.	1. Select Login 2. Enter email: ali.khan@test.com 3. Enter password: WrongPass!1 4. Observe response	Email: ali.khan@test.com Password: WrongPass!1	System displays 'Email or password does not match with user credentials!'		
TC-UA-011	Login with Non-Existent Email (Negative)	No account exists for unknown@test.com.	1. Select Login 2. Enter email: unknown@test.com 3. Enter password: Ali@12345 4. Observe response	Email: unknown@test.com Password: Ali@12345	System displays 'Email or password does not match with user credentials!'		
TC-UA-012	View Account Details – Valid Credentials (Positive)	User account exists.	1. Select option 3 (Check Account Details) 2. Enter email: ali.khan@test.com 3. Enter password: Ali@12345 4. Observe output	Email: ali.khan@test.com Password: Ali@12345	System displays first name, last name, CNIC, phone number, email, and password.		
TC-UA	View Account Details – Invalid	User account exists.	1. Select option 3 2. Enter email:	Email: wrong@test.com	System displays		

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
-013	Credentials (Negative)		wrong@test.com 3. Enter password: Ali@12345 4. Observe output		'Email or password does not match with user credentials!'		
TC-UA-014	Edit Account Details – Successful Update (Positive)	Logged-in user account exists.	1. Select option 4 (Edit Account Details) 2. Enter correct email and password 3. Enter new first name: Ahmed 4. Enter new last name: Ali 5. Enter new phone: 03219876543 6. Enter new valid password: Ahmed@5678 7. Observe output	New Name: Ahmed Ali New Phone: 03219876543 New Pass: Ahmed@5678	System displays 'Details changed successfully!'		
TC-UA-015	Edit Account Details – Wrong Credentials (Negative)	User account exists.	1. Select option 4 2. Enter wrong email or password 3. Observe response	Email: wrong@test.com	System displays 'Email or password does not match with user credentials!'		
TC-UA-016	Forgot Password – Valid Email and Phone (Positive)	User account with matching email and phone exists.	1. Select option 5 (Forgot Password) 2. Enter email: ali.khan@test.com 3. Enter phone: 03001234567 4. Enter 4-digit code: 1234 5. Enter new password: NewPass@9 6. Observe output	Email: ali.khan@test.com Phone: 03001234567 4-Digit Code: 1234 New Pass: NewPass@9	System displays 'Password Changed Successfully!'		
TC-UA-017	Forgot Password – Phone Number Not Matching (Negative)	User account exists.	1. Select option 5 2. Enter email: ali.khan@test.com 3. Enter wrong phone: 09999999999 4. Observe response	Email: ali.khan@test.com Phone: 09999999999	System displays 'Phone number or email does not exist in Database!'		
TC-UA-018	Forgot Password – Code Less Than 4 Digits (Boundary/Negative)	Valid email and phone combination entered.	1. Select option 5 2. Enter valid email and phone 3. Enter 3-digit code: 123 4. Observe response	Code: 123	System displays 'Password must have 4 digits!' and re-prompts for code.		

9.2 Module 2 – Flight Search

This module covers flight search functionality including airport selection, country/airport input, date validation, and flight type selection.

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-FS-001	Search One-Way Flight from Lahore to Dubai (DIA) (Positive)	System is running.	1. Select option 6 (Search Flight) 2. Select departure airport: 1 (Lahore) 3. Enter arrival country: UAE 4. Enter arrival airport: DIA 5. Select flight type: 1 (One-Way) 6. Enter date: 15/06/2026 7. Observe output	Departure: Lahore Country: UAE Airport: DIA Date: 15/06/2026 Type: One-Way	System displays flight ID 'LHE-DXB-30T1R' with flight name 'PIA Express', price 100000, departure 08:30, arrival 10:00.		
TC-FS-002	Search Return Flight from Lahore to Dubai (DIA) (Positive)	System is running.	1. Select Search Flight 2. Departure: 1 (Lahore) 3. Country: UAE 4. Airport: DIA 5. Type: 2 (Return) 6. Date: 15/06/2026	Departure: Lahore Country: UAE Airport: DIA Date: 15/06/2026 Type: Return	System displays flight ID 'LHE-DXB-301R' with return date 17/06/2026, price 100000, return time 11:30 GMT+4.		
TC-FS-003	Search One-Way Flight from Faisalabad to Saudi Arabia (KIA) (Positive)	System is running.	1. Departure: 2 (Faisalabad) 2. Country: Saudi Arabia 3. Airport: KIA 4. Type: One-Way 5. Date: 20/07/2026	Departure: Faisalabad Country: Saudi Arabia Airport: KIA Date: 20/07/2026	System displays flight ID 'FIA-KIA-12L09F' with price 150000.		
TC-FS-004	Search One-Way Flight from Islamabad to Sharjah (SIA) (Positive)	System is running.	1. Departure: 3 (Islamabad) 2. Country: UAE 3. Airport: SIA 4. Type:	Departure: Islamabad Country: UAE Airport: SIA Date: 10/08/2026	System displays flight ID 'ISB-SIA-X201F' with flight name 'AirBlue Sky', price 100000.		

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
			One-Way 5. Date: 10/08/2026				
TC-FS-005	Enter Invalid Country Name (Negative)	System is running.	1. Select Search Flight 2. Enter arrival country: India 3. Observe response	Country: India	System displays 'You Entered Wrong Details!' and re-prompts for arrival country.		
TC-FS-006	Enter Invalid Airport Code (Negative)	Valid country (UAE) selected.	1. Select UAE as country 2. Enter arrival airport: XYZ 3. Observe response	Airport Code: XYZ	System displays 'Please Enter Valid input!' and re-prompts for airport.		
TC-FS-007	Enter Past Date (Boundary/Negative)	System is running.	1. Select Search Flight with valid airport details 2. Enter departure date: 01/01/2024 3. Observe response	Date: 01/01/2024	System displays 'Please input valid Date!' and re-prompts.		
TC-FS-008	Enter Today's Date (Boundary/Positive)	System is running.	1. Select Search Flight with valid airport details 2. Enter today's date: 30/03/2026 3. Observe response	Date: 30/03/2026	System accepts today's date and searches for flights.		
TC-FS-009	Enter Date with Invalid Month (Negative)	System is running.	1. Select Search Flight 2. Enter date: 15/13/2026 (month 13) 3. Observe response	Date: 15/13/2026	System displays 'Please input valid Date!' and re-prompts.		
TC-FS-010	Enter Date with Invalid Day (Negative)	System is running.	1. Select Search Flight 2. Enter date: 32/06/2026 3. Observe response	Date: 32/06/2026	System displays 'Please input valid Date!' and re-prompts.		

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-FS-011	Select Invalid Departure Airport Number (Negative)	System is running.	1. Select Search Flight 2. Enter departure airport: 5 3. Observe response	Departure Selection: 5	System displays 'You Entered Wrong Details!' and re-prompts for departure airport.		
TC-FS-012	Flight Search Using Airport Number Instead of Code (Positive)	System is running.	1. Departure: 1 (Lahore) 2. Country: UAE 3. Enter airport as: 1 (DIA equivalent) 4. Type: One-Way 5. Date: 15/06/2026	Airport Input: '1' (numeric equivalent of DIA)	System correctly maps input '1' to DIA and displays the corresponding flight.		
TC-FS-013	Verify All UAE Airports from All Departure Cities (Positive)	System is running.	1. Iterate all 3 departure airports 2. Select UAE country 3. Select each of DIA, SIA, AIA in turn 4. Verify flight ID returned	All 9 combinations: Lahore/Faisalabad/Islamabad × DIA/SIA/AIA	Each combination returns a unique non-empty flight ID.		
TC-FS-014	Verify All Saudi Arabia Airports from All Departure Cities (Positive)	System is running.	1. Iterate all 3 departure airports 2. Select Saudi Arabia country 3. Select each of KIA, FIA, AIA in turn	All 9 combinations: Lahore/Faisalabad/Islamabad × KIA/FIA/AIA	Each combination returns a unique non-empty flight ID.		
TC-FS-015	Verify Return Flight Has Return Date and Time (Positive)	Return flight search performed.	1. Select any departure airport 2. Select UAE, DIA 3. Choose Return type 4. Enter date: 01/05/2026	Type: Return Date: 01/05/2026	Flight information includes 'Return Date' and 'Return Time' fields in the output.		
TC-FS-016	Verify One-Way Flight Has No Return Date (Positive)	One-Way flight search performed.	1. Select any departure airport 2. Select UAE, DIA	Type: One-Way	Flight information does NOT include 'Return Date'		

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
			3. Choose One-Way type 4. Enter date: 01/05/2026		or 'Return Time' fields.		
TC-FS-017	Non-Numeric Input for Departure Airport Selection (Negative)	System is running.	1. When prompted for departure airport (1,2,3) 2. Enter text: 'abc' 3. Observe response	Input: abc	System displays 'Enter valid Input!' (cin error handling), clears buffer, and re-prompts.		

9.3 Module 3 – Seat Management

This module covers seat booking, availability display, class-range enforcement, seat cancellation, and double-booking prevention.

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-SM-001	Book Economy Seat Successfully (Positive)	Valid flight selected; system in booking flow.	1. Enter passenger details 2. Select class: 1 (Economy) 3. View available economy seats (1-30) 4. Enter seat number: 5 5. Observe response	Class: Economy Seat: 5	System displays 'Seat selected successfully!' and records seat 5 as booked.		
TC-SM-002	Book Business Class Seat Successfully (Positive)	Valid flight selected; system in booking flow.	1. Select class: 2 (Business) 2. View available business seats (31-40) 3. Enter seat number: 35 4. Observe response	Class: Business Seat: 35	System displays 'Seat selected successfully!' and records seat 35 as booked.		
TC-SM-003	Book First Class Seat Successfully (Positive)	Valid flight selected; system in booking flow.	1. Select class: 3 (First Class) 2. View available first-class seats (41-50) 3. Enter seat number: 45 4. Observe response	Class: First Class Seat: 45	System displays 'Seat selected successfully!' and records seat 45 as booked.		
TC-SM-004	Attempt to Book Already-Booked Seat (Negative)	Seat 5 (Economy) already booked in current session.	1. Select class: 1 (Economy) 2. Enter seat number: 5 (already booked) 3. Observe response	Seat: 5 (booked)	System displays 'Seat already Booked!' and re-prompts for another seat.		
TC-SM-005	Enter Seat Number Out of Class Range (Negative)	Economy class selected (valid range 1-30).	1. Select class: 1 (Economy) 2. Enter seat number: 35 (Business range) 3. Observe response	Class: Economy Seat: 35	System displays 'Invalid seat number for this class!' and re-prompts.		
TC-SM-006	Enter Seat Number 0 (Boundary/Negative)	Any class selected.	1. Select class 2. Enter seat number: 0 3. Observe response	Seat: 0	System handles invalid input; for cancel_seat: displays 'Please Select a valid seat!'. For book_seat: 'Invalid seat number for this class!'		
TC-SM-007	Enter Seat Number 51 (Out of Range, Boundary/Negative)	Any class selected.	1. Select class 2. Enter seat number: 51 3.	Seat: 51	System detects out-of-range (>50) and displays		

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
			Observe response		appropriate error.		
TC-SM-008	Cancel a Booked Seat Successfully (Positive)	Seat 5 booked for current user in this session.	1. Select option 8 (Cancel Flight) 2. Enter flight ID 3. Enter flight date 4. Enter correct CNIC 5. Confirm cancellation: Yes 6. Enter seat number: 5 7. Observe response	Flight ID: valid CNIC: matching Seat: 5	System displays 'Seat Cancelled successfully!' and marks seat 5 as available.		
TC-SM-009	Attempt to Cancel Someone Else's Seat (Negative)	Seat 10 booked by a different user.	1. Select cancel flow 2. Enter CNIC matching current user 3. Enter seat: 10 (belongs to another user) 4. Observe response	CNIC: current user Seat: 10 (other user's)	System displays 'This is not your seat!' and re-prompts.		
TC-SM-010	Double Cancellation of Same Seat (Negative)	Seat 5 already cancelled (marked available).	1. Attempt to cancel seat 5 again 2. Observe response	Seat: 5 (already free)	System displays 'Seat is already Freed!' and re-prompts.		
TC-SM-011	View Available Economy Seats (Positive)	Flight with known booked seats exists.	1. Reach class selection in booking 2. Select Economy class 3. Observe seat display	Class: Economy	System displays all unbooked seat numbers in range 1-30. Booked seats are not shown.		
TC-SM-012	View Available Business Seats (Positive)	Flight with known booked seats exists.	1. Select Business class 2. Observe seat display	Class: Business	System displays all unbooked seat numbers in range 31-40.		
TC-SM-013	View Available First Class Seats (Positive)	Flight selected.	1. Select First Class 2. Observe seat display	Class: First Class	System displays all unbooked seat numbers in range 41-50.		
TC-SM-014	Seat Availability Updates After Booking (Positive)	All 30 economy seats initially available.	1. Book seat 7 (Economy) 2. Re-display available economy seats 3. Verify seat 7 not shown	Seat booked: 7	Seat 7 is absent from the available seats list after booking.		
TC-SM-015	Seat Availability Updates After Cancellation (Positive)	Seat 7 (Economy) previously booked.	1. Cancel seat 7 2. Re-display available economy seats 3. Verify seat 7 appears	Seat cancelled: 7	Seat 7 reappears in the available seats list after cancellation.		

9.4 Module 4 – Booking & Passenger Details

This module covers passenger count/type validation, baggage weight boundaries, and fare calculation logic.

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-BP-001	Booking with 1 Adult Passenger (Positive)	User is authenticated (logged in or signed up).	1. Start booking flow 2. Enter no. of passengers: 1 3. Enter adults: 1 4. Enter passenger details (valid name, CNIC, passport, phone) 5. Select Economy class, seat 3 6. Enter baggage weight: 10kg 7. Observe flow	Passengers: 1 Adult Seat: 3 Baggage: 10kg	Booking progresses to payment; fare displayed as 100000 + 10*1000 = 110000 PKR for UAE.		
TC-BP-002	Booking with Maximum 5 Passengers (Boundary/Positive)	User is authenticated.	1. Enter no. of passengers: 5 2. Enter adults: 3, children: 1, infants: 1 3. Enter all passenger details 4. Observe flow	Passengers: 5 (3A+1C+1I)	System accepts 5 passengers, prompts details for each, and proceeds to payment.		
TC-BP-003	Booking with 0 Passengers (Boundary/Negative)	User is authenticated.	1. Enter no. of passengers: 0 2. Observe response	Passengers: 0	System displays 'PLease give valid input!' and re-prompts.		
TC-BP-004	Booking with More Than 5 Passengers (Boundary/Negative)	User is authenticated.	1. Enter no. of passengers: 6 2. Observe response	Passengers: 6	System displays 'PLease give valid input!' (exceeds limit of 5).		
TC-BP-005	Child Booking Without Adult (Negative)	User is authenticated.	1. Enter passengers: 1 2. Enter adults: 0 3. Enter children: 1 4. Observe response	Adults: 0 Children: 1	System displays 'Child cant travel without An Adult!' and re-prompts.		
TC-BP-006	Infant Booking Without Adult (Negative)	User is authenticated.	1. Enter passengers: 1 2. Enter adults: 0 3. Enter infants: 1 4. Observe response	Adults: 0 Infants: 1	System displays 'Infants cant travel without An Adult!' and re-prompts.		
TC-BP-007	Adults+Children Exceeding Total Passenger Count (Negative)	Total passengers set to 3.	1. Enter passengers: 3 2. Enter adults: 3 3. Enter children: 1 4.	Total: 3, Adults: 3, Children: 1	System detects sum exceeds total (3+1=4 > 3) and displays		

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
			Observe response		'Maximum Limit Reached!'		
TC-BP-008	Enter Baggage Weight of 0kg (Boundary/Positive)	Booking flow active; class and seat selected.	1. Enter baggage weight: 0 2. Observe fare calculation	Baggage: 0kg	System accepts 0kg weight. Fare = base price + 0 = base price only.		
TC-BP-009	Enter Baggage Weight of 20kg (Boundary/Positive)	Booking flow active.	1. Enter baggage weight: 20 2. Observe fare calculation for UAE	Baggage: 20kg	System accepts 20kg. Fare = 100000 + (20 * 1000) = 120000 PKR.		
TC-BP-010	Enter Baggage Weight Exceeding 20kg (Boundary/Negative)	Booking flow active.	1. Enter baggage weight: 25 2. Observe response	Baggage: 25kg	System displays 'Weight should be less than 20KG!' and re-prompts.		
TC-BP-011	Enter Negative Baggage Weight (Negative)	Booking flow active.	1. Enter baggage weight: -5 2. Observe response	Baggage: -5	System displays 'Weight should be less than 20KG!' and re-prompts.		
TC-BP-012	Fare Calculation for Saudi Arabia Flight (Positive)	Saudi Arabia flight selected.	1. Select SA flight 2. Enter baggage: 15kg 3. Proceed to payment	Country: Saudi Arabia Baggage: 15kg	Fare = 150000 + (15 * 1000) = 165000 PKR.		

9.5 Module 5 – Payment

This module covers all payment methods, all 5 supported currencies, currency conversion, and payment sufficiency checking.

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-PM-001	Successful Payment via Debit Card in PKR (Positive)	Booking completed, total fare displayed.	1. Enter 'Yes' to book 2. Select currency: any (amount enters as PKR equivalent) 3. Select payment method: 1 (Debit Card) 4. Enter card details (name, 16-digit number, expiry, CCV) 5. Enter amount equal to total fare 6. Observe response	Payment: exact fare amount Method: Debit Card	System displays 'Flight Booked Successfully'.		
TC-PM-002	Payment with Amount Greater than Fare (Positive)	Booking completed.	1. Enter payment amount: fare + 5000 2. Observe response	Payment: fare + 5000 (excess)	System displays 'We will send back your remaining amount soon = 5000' and then 'Flight Booked Successfully'.		
TC-PM-003	Payment with Insufficient Amount (Negative)	Booking completed.	1. Enter payment amount less than total fare 2. Observe response	Payment: fare - 10000 (insufficient)	System displays 'Payment Insufficient' and re-prompts for amount.		
TC-PM-004	Currency Conversion – USD to PKR (Positive)	On payment step, total fare = 110000 PKR.	1. Select currency: 5 (USD, rate=285) 2. Enter amount in USD: 386.0 (approx 110000/285) 3. Observe if booking succeeds	Currency: USD Amount: 400 USD (=114000 PKR)	System converts 400 USD to 400*285 = 114000 PKR ≥ fare; booking succeeds.		
TC-PM-005	Currency Conversion – UAE Dirham (Positive)	Total fare = 100000 PKR.	1. Select currency: 1 (Dirham, rate=76.37) 2. Enter amount: 1311 Dirhams (≈100082 PKR) 3. Observe response	Currency: Dirham Amount: 1311	System converts 1311 × 76.37 ≈ 100122 PKR ≥ fare; booking succeeds.		
TC-PM-006	Currency Conversion – EUR (Positive)	Total fare = 100000 PKR.	1. Select currency: 2 (EUR, rate=310) 2. Enter amount: 323 EUR (=100130 PKR) 3. Observe response	Currency: EUR Amount: 323	System converts 323 × 310 = 100130 PKR ≥ fare; booking succeeds.		
TC-PM-007	Currency Conversion – British Pound (Positive)	Total fare = 100000 PKR.	1. Select currency: 3 (Pound, rate=360) 2. Enter amount: 278 Pounds (=100080 PKR)	Currency: Pound Amount: 278	System converts 278 × 360 = 100080 PKR ≥ fare; booking succeeds.		

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-PM-008	Currency Conversion – Saudi Riyal (Positive)	Total fare = 150000 PKR.	1. Select currency: 4 (SAR, rate=74.79) 2. Enter amount: 2007 SAR (≈150063 PKR)	Currency: SAR Amount: 2007	System converts 2007 × 74.79 ≈ 150084 PKR ≥ fare; booking succeeds.		
TC-PM-009	Invalid Currency Choice (Negative)	On currency selection step.	1. Enter currency choice: 0 or 6 2. Observe response	Currency choice: 0 or 6	System displays 'Invalid choice!' and re-prompts for currency.		
TC-PM-010	Payment via Online Banking (Positive)	Booking completed.	1. Select payment method: 3 (Online Banking) 2. Enter bank name, account number, receiver account, password 3. Enter sufficient amount 4. Observe response	Method: Online Banking Amount: sufficient	System displays 'Flight Booked Successfully'.		
TC-PM-011	Invalid Payment Method Selection (Negative)	On payment method selection.	1. Enter payment method: 4 2. Observe response	Method choice: 4	System displays 'Invalid Choice!' and re-prompts.		
TC-PM-012	Decline Booking Before Payment (Positive)	Flight found and displayed.	1. System asks 'Do you want to book this flight' 2. Enter: 2 (No) 3. Observe response	Decision: No	System displays 'OK No Problem Thanks for visiting!' and returns to menu.		
TC-PM-013	Debit Card with Invalid Expiry Date (Negative)	Debit card payment method selected.	1. Enter card details 2. Enter expiry date: 01/01/2024 (past date) 3. Observe response	Expiry Date: 01/01/2024	System displays 'Please input valid Date!' and re-prompts for expiry date.		

9.6 Module 6 – Receipt & Flight Cancellation

This module covers receipt completeness and the flight cancellation flow including CNIC matching and seat release.

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-RC-001	Generate Receipt After Successful Booking (Positive)	Booking and payment completed successfully.	1. Complete full booking + payment flow 2. Observe receipt output after payment	Booking: completed with all passenger details	System displays 'Here is your Ticket Receipt:' followed by all passenger details and flight information including total paid amount.		
TC-RC-002	Receipt Displays All Passenger Details (Positive)	Booking with 2 passengers completed.	1. Complete booking with 2 passengers 2. Review receipt output	Passengers: 2	Receipt shows Passenger Details 1 and Passenger Details 2, each with first name, last name, CNIC, passport, and phone number.		
TC-RC-003	Receipt Displays Correct Flight Information (Positive)	Booking for Lahore → Dubai (DIA), One-Way completed.	1. Complete booking for LHE→DIA One-Way 2. Review receipt	Flight: LHE-DXB One-Way	Receipt shows flight ID, flight name, type, from, to, date, price, baggage allowance, departure/arrival times.		
TC-RC-004	Cancel Flight with Correct CNIC (Positive)	Flight booked and payment recorded.	1. Select option 8 (Cancel Flight) 2. Enter correct flight ID 3. Enter correct flight date 4. Enter correct CNIC 5. Confirm: 1 (Yes) 6. Enter booked seat number 7. Observe response	CNIC: matching booked user	System displays 'Flight cancelled Successfully!' and frees the seat.		
TC-RC-005	Cancel Flight with Wrong CNIC (Negative)	Flight booked for user with known CNIC.	1. Select option 8 2. Enter valid flight ID and date 3. Enter wrong CNIC: 9999999999999 4. Observe response	CNIC: 9999999999999 (wrong)	System displays 'CNIC does not match!' without cancelling the flight.		
TC-RC-006	Cancel Flight – User Declines Confirmation (Positive)	Flight booked; CNIC matches.	1. Select option 8 2. Enter correct CNIC 3. When prompted, enter: 2 (No) 4. Observe response	Confirmation: No	System displays 'OK As you Wish!' and does not cancel the booking.		
TC-RC-007	Cancel Non-Existent Flight	No booking exists for the entered CNIC.	1. Select option 8 2. Enter CNIC not associated with any booking 3.	CNIC: no booking exists	System displays 'CNIC does not match!'		

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
	Booking (Negative)		Observe response				
TC-RC-008	Enter Invalid Date for Cancellation (Negative)	Cancel flight flow initiated.	1. Select option 8 2. Enter flight ID 3. Enter date: 32/13/2025 4. Observe response	Date: 32/13/2025	System displays 'Please input valid Date!' and re-prompts.		

9.7 Module 7 – Input Validation

This module covers system-wide input handling for non-numeric inputs, invalid menu selections, and edge-case string inputs.

TC ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-IV-001	Non-Numeric Input for Integer Fields (Negative)	System prompts for CNIC.	1. When prompted for CNIC, enter: 'abc' 2. Observe response	Input: abc (for numeric field)	System clears cin buffer and displays 'Enter valid CNIC!' and re-prompts.		
TC-IV-002	Non-Numeric Input for Phone Number (Negative)	System prompts for phone number.	1. Enter phone: 'phone' 2. Observe response	Phone: phone (non-numeric)	System displays 'Enter valid Phone Number!' and re-prompts.		
TC-IV-003	Non-Numeric Input for Passport (Negative)	System prompts for passport.	1. Enter passport: 'ABCD' 2. Observe response	Passport: ABCD	System displays 'Enter valid Passport!' and re-prompts.		
TC-IV-004	System Handles Invalid Menu Choice Gracefully (Negative)	Main menu displayed.	1. Enter choice: 99 2. Observe response	Menu choice: 99	System displays 'You Entered Wrong Choice!' and redisplays the main menu.		
TC-IV-005	Empty First Name (Negative)	Sign-up flow active.	1. When prompted for first name, press Enter (empty input) 2. Observe response	First Name: (empty)	System validates: empty name passes name check but re-prompt may occur based on any further validation flow.		
TC-IV-006	Password with Only Spaces (Negative)	Sign-up password entry.	1. Enter password: ' ' (8 spaces) 2. Observe response	Password: 8 spaces	System detects no uppercase, no digit, no special char → displays 'Password does not match with criteria'.		
TC-IV-007	Exit System (Positive)	Main menu displayed.	1. Enter choice: 9 2. Observe response	Choice: 9 (Exit)	System exits the main loop and program terminates gracefully.		

10. Traceability Matrix

The following traceability matrix maps requirements from the SRS to their corresponding use cases and test cases, ensuring forward and backward traceability.

Sr.	Requirement ID	Use Case ID	GUI ID	Test Case ID	Description	Status	Date Completed
1	FR-UA-1.1: Sign-Up	UC-01	GUI-01	TC-UA-001 to TC-UA-008	User account creation with validation		
2	FR-UA-1.2: Login	UC-02	GUI-02	TC-UA-009 to TC-UA-011	User login with credential verification		
3	FR-UA-1.3: View Details	UC-03	GUI-03	TC-UA-012 to TC-UA-013	View account details		
4	FR-UA-1.4: Edit Details	UC-04	GUI-04	TC-UA-014 to TC-UA-015	Edit profile details		
5	FR-UA-1.5: Forgot Password	UC-05	GUI-05	TC-UA-016 to TC-UA-018	Password recovery via email/phone		
6	FR-FS-2.1: Search Flights	UC-06	GUI-06	TC-FS-001 to TC-FS-017	Flight search by route, date, and type		
7	FR-BP-3.1: Passenger Details	UC-07	GUI-07	TC-BP-001 to TC-BP-012	Passenger count and type validation		
8	FR-SM-3.2: Seat Selection	UC-08	GUI-08	TC-SM-001 to TC-SM-015	Seat booking, availability, cancellation		
9	FR-PM-3.3: Payment	UC-09	GUI-09	TC-PM-001 to TC-PM-013	Payment processing and currency conversion		
10	FR-RC-4.1: Receipt Generation	UC-10	GUI-10	TC-RC-001 to TC-RC-003	Receipt display after booking		
11	FR-CF-5.1: Cancel Flight	UC-11	GUI-11	TC-RC-004 to TC-RC-008	Flight cancellation by CNIC		
12	NFR-REL-4.1: Input Validation	UC-12	N/A	TC-IV-001 to TC-IV-007	Graceful handling of invalid inputs		

11. Major Deliverables

The following items will be delivered at the completion of the testing phase:

- Test Plan Document (this document) – ATRS-2025-TP
- Complete Test Case Document with 90 test cases across 7 modules
- Defect Log – documenting all defects found with severity, steps, and status
- Test Execution Report – recording actual results and pass/fail status for each test case
- Test Summary Report – overall testing summary including pass/fail statistics, coverage analysis, and recommendations
- Traceability Matrix – mapping requirements to test cases

12. Risks and Assumptions

No.	Risk / Assumption	Probability	Impact	Mitigation
1	Incomplete or changing requirements	Medium	High	Freeze requirements before testing begins
2	Insufficient test data for all routes	Medium	Medium	Generate synthetic test data covering all airport/country combos
3	Tight testing schedule	High	Medium	Prioritize critical path test cases; use parallel execution
4	Non-availability of testing environment	Low	High	Set up dedicated test machine with compiled C++ binary
5	Lack of automated testing tools	High	Low	Perform manual testing with structured checklists
6	Code defects causing crashes	Medium	High	Conduct unit testing before integration testing
7	Payment module not fully implemented	Low	High	Mock payment inputs; test validation logic only
8	In-memory data lost between sessions	High	Medium	Note limitation; test within single session runs

13. Exit Criteria

The following conditions must be met before the testing phase is considered complete:

- All 90 test cases have been executed with a documented result (Pass / Fail / Not Executed).
- Zero Critical severity defects remain open.
- No more than two Major severity defects remain open (with documented justification).
- All functional requirements from SRS v2.0 are covered by at least one passing test case.
- All core user journeys (Sign-Up → Search → Book → Pay → Receipt; and Cancel Flow) execute without system crashes.
- Input validation test cases (TC-IV-001 to TC-IV-007) all pass.
- Test Summary Report has been reviewed and approved by the QA Lead.

14. References

Ref. No.	Document Title	Date	Source
ATRS-2025-SRS	Software Requirements Specification v2.0	2025	Project Repository / Team Documentation
ATRS-2025-SRS1	Software Requirements Specification v1.0	2025	Project Repository / Team Documentation
ATRS-2025-TC	Air Ticket Reservation Test Cases (Draft)	2025	QA Documentation Folder
ATRS-2025-CODE	Air Ticket Reservation C++ Source Code	2025	Project Repository
ATRS-2025-TP-T	Test Plan Template	2003	FAST-NU / Academic Repository

15. Appendices

Appendix A – Test Case Summary Count

No.	Module	Total TCs	Positive	Negative/BV
1	Module 1 – User Authentication	18	8	10
2	Module 2 – Flight Search	17	8	9
3	Module 3 – Seat Management	15	8	7
4	Module 4 – Booking & Passenger Details	12	5	7
5	Module 5 – Payment	13	9	4
6	Module 6 – Receipt & Cancellation	8	5	3
7	Module 7 – Input Validation	7	1	6
	TOTAL	**90**	**44**	**46**

Appendix B – Flight Route Reference

The following table summarises the 54 available route combinations (3 departure airports × 2 countries × 3 airports × 2 flight types = 36 routes; plus return flights = 54 total flight IDs) implemented in the source code:

Departure	Country	Airport	Type	Flight ID
Lahore	UAE	DIA	One-Way	LHE-DXB-30T1R
Lahore	UAE	DIA	Return	LHE-DXB-301R
Lahore	UAE	SIA	One-Way	LHE-SIA-20E1F
Lahore	UAE	SIA	Return	LHE-SIA-201F
Lahore	UAE	AIA	One-Way	LHE-AIA-30Y1R
Lahore	UAE	AIA	Return	LHE-AIA-101A
Lahore	Saudi Arabia	KIA	One-Way	LHE-KIA-12B09F
Lahore	Saudi Arabia	KIA	Return	LHE-KIA-409F
Lahore	Saudi Arabia	FIA	One-Way	LHE-FIA-A2701F
Lahore	Saudi Arabia	FIA	Return	LHE-FIA-1C01A
Lahore	Saudi Arabia	AIA	One-Way	LHE-AIA-BF301R
Lahore	Saudi Arabia	AIA	Return	LHE-AIA-30S1R
Faisalabad	UAE	DIA	One-Way	FIA-DXB-W6301R
Faisalabad	UAE	DIA	Return	FIA-DXB-N3501R
Islamabad	UAE	DIA	One-Way	ISB-DXB-6301R0
Islamabad	UAE	DIA	Return	ISB-DXB-3501RZ

(Refer to source code Flights class for the complete 54-entry reference table.)

Appendix C – Currency Conversion Reference

Choice	Currency	Rate (PKR)	Example: 100,000 PKR
1	UAE Dirham (AED)	76.37	≈ 1,309 AED
2	Euro (EUR)	310.00	≈ 323 EUR
3	British Pound (GBP)	360.00	≈ 278 GBP
4	Saudi Riyal (SAR)	74.79	≈ 1,337 SAR
5	US Dollar (USD)	285.00	≈ 351 USD